37.3: Plant nutrition often involves relationships with other organisms

To this point, we have looked at plants as exploiters of soil resources, but plants and soil have a two-way relationship. Dead plants provide much of the energy needed by soil-dwelling bacteria and fungi. Many of these organisms also benefit from sugar-rich secretions produced by living roots. Meanwhile, plants derive benefits from their associations with soil bacteria and fungi. As shown in **Figure 37.9**, mutually beneficial relationships across kingdoms and domains are not rare in nature. However, they are of particular importance to plants. We'll explore some important *mutualisms* between plants and soil bacteria and fungi, as well as some unusual, nonmutualistic forms of plant nutrition.

Figure 37.9

AP Make Connections: Mutualism Across Kingdoms and Domains

The figure gives examples of mutualism between fungus-bacterium, animal-bacterium, plant-bacterium, animal-fungus, plant-fungus, and plant-animal.

Mutualism across kingdoms and domains:

Some toxic species of fish don't make their own poison. How is that possible? Some species of ants chew leaves but don't eat them. Why? The answers lie in some amazing mutualisms, relationships between different species in which each species provides a substance or service that benefits the other (see Concept 54.1). Sometimes mutualisms occur within the same kingdom, such as between two species of animals. Many mutualisms, however, involve species from different kingdoms or domains, as in these examples.

Fungus-bacterium: A lichen is a mutualistic association between a fungus and a photosynthetic partner. In the lichen Peltigera, the photosynthetic partner is a species of cyanobacterium. The cyanobacterium supplies carbohydrates, while the fungus provides anchorage, protection, minerals, and water. (See figure 31.22). Two photographs show the purplish lichen Peltigera and a micrograph of a longitudinal section of Peltigera showing green photosynthetic bacteria sandwiched between layers of fungus.

Animal-bacterium: Fugu is the Japanese name for puffer fish and the delicacy made from it, which can be deadly. Most species of puffer fish contain lethal amounts of the nerve toxin tetrodotoxin in their organs, especially the liver, ovaries, and intestines. Therefore, a specially trained chef must remove the poisonous parts. The tetrodotoxin is synthesized by mutualistic bacteria (various Vibrio species) associated with the fish. The fish gains a potent chemical defense, while the bacteria live in a high-nutrient, low-competition environment. A photograph shows a puffer fish (fugu).

Plant-bacterium: The floating fern Azolla provides carbohydrates for a nitrogen-fixing cyanobacterium that resides in the air spaces of the leaves. In return, the fern receives nitrogen from the cyanobacterium. (See concept 27.5). A photograph shows the fern Azolla floating in water.

Animal-fungus: Leaf-cutter ants harvest leaves that they carry back to their nest, but the ants do not eat the leaves. Instead, a fungus grows by absorbing nutrients from the leaves, and the ants eat part of the fungus that they have cultivated. Two photographs show leaf-cutter ants bringing leaves to their nest, and a group of ants tending a fungal garden in their nest.

Plant-fungus: Most plant species have mycorrhizae, mutualistic associations between roots and fungi. The fungus absorbs carbohydrates from the roots. In return, the fungus's mycelium, a dense network of filaments called hyphae, increases the surface area for the uptake of water and minerals by roots. (See figure 31.4). An SEM shows a fungus growing on the root of a sorghum plant.

Plant-animal: Some species of acacia plants are aggressively defended from predators and competitors by ants that live within the plant's hollow thorns. The plant provides nourishment for the ants in the forms of protein-rich structures at the bases of leaves and carbohydrate-rich nectar. (See figure 54.8). A photograph shows protective ants harvesting protein-rich structures from an acacia plant.

MAKE CONNECTIONS *Describe three more examples of mutualisms. (See*Figure 27.20*,***Figure 38.4**, and**Concept 41.4**.)*

*For suggested answer, see*Appendix A.*

Bacteria and Plant Nutrition

A variety of soil bacteria play roles in plant nutrition. Some engage in mutually beneficial chemical exchanges with plant roots. Others enhance the decomposition of organic materials and increase nutrient availability.

Rhizobacteria

Rhizobacteria are bacteria that live either in close association with plant roots or in the **rhizosphere**, the soil closely surrounding plant roots. Many rhizobacteria form mutually beneficial associations with plant roots. Rhizobacteria depend on nutrients such as sugars, amino acids, and organic acids that are secreted by plant cells. Up to 20% of a plant's photosynthetic production may be used to fuel these complex bacterial communities. In return, plants reap many benefits from these mutualistic associations. Some rhizobacteria produce antibiotics that protect roots from disease. Others absorb toxic metals or make nutrients more available to roots. Still others convert gaseous nitrogen into forms usable by the plant or produce chemicals

that stimulate plant growth. Inoculation of seeds with plant-growth-promoting rhizobacteria can increase crop yield and reduce the need for fertilizers and pesticides.

Some rhizobacteria are free-living in the rhizosphere, whereas others are **endophytes** that live between cells within the plant. Both the intercellular spaces occupied by endophytic bacteria and the rhizosphere associated with each plant root system contain a unique and complex cocktail of root secretions and microbial products that differ from those of the surrounding soil. A recent study revealed that the compositions of bacterial communities living endophytically and in the rhizosphere are not identical **(Figure 37.10)**. A better understanding of the types of bacteria within and around roots could potentially have profound agricultural benefits.

Figure 37.10

Inquiry: How variable are the compositions of bacterial communities inside and outside of roots?

Experiment

The bacterial communities found within and immediately outside of root systems are known to improve plant growth. In order to devise agricultural strategies to increase the benefits of these bacterial communities, it is necessary to determine how complex they are and what factors affect their composition. A problem with studying these bacterial communities is that a handful of soil contains as many as 10,000 types of bacteria, more than all the bacterial species that have been described. Researchers cannot simply culture each species and use a taxonomic key to identify them; a molecular approach is needed.

A fluorescent LM micrograph of green bacteria growing on the surface of a root.

Bacteria (green) on surface of root (fluorescent LM)

Jeffery Dangl and his colleagues estimated the number of bacterial “species” in samples using a technique called *metagenomics* (see Concept 21.1). The bacterial community samples differed in location (endophytic, inside the rhizosphere, or outside the rhizosphere), soil type (having large amounts of clay or porous), and the developmental stage of the root system with which they were associated (old or young). The DNA from each sample was purified, and the polymerase chain reaction (PCR) was used to amplify the DNA that codes for the 16S ribosomal RNA subunits. Many thousands of DNA sequence variations were found in each sample. The researchers lumped the sequences that were more than 97% identical into “taxonomic units” or “species.” (The word *species* is in quotation marks because “two organisms having a single gene that is more

than 97% identical" is not explicit in any definition of species.) Having established the types of "species" in each community, the researchers made a tree diagram showing the percent of bacterial "species" found in common in each community.

Results

This tree diagram breaks down the relatedness of bacterial communities into finer and finer levels of detail. The two explanatory labels give examples of how to interpret the diagram.

A tree diagram of the relatedness of bacterial communities.

Bacteria inside roots in clayey soil and bacteria inside roots in porous soil are 34% similar. Soil bacteria inside younger and inside older roots outside the rhizosphere in porous soil are 80% similar.

Data from D. S. Lundberg et al., Defining the core *Arabidopsis thaliana* root microbiome, *Nature* 488:86–94 (2012).

Conclusion

The "species" composition of the bacterial communities varied markedly according to the location inside the root versus outside the root and according to soil type.

INTERPRET THE DATA

- *Which of the three community locations was least like the other two?*
- *Rank the three variables (community location, developmental stage of roots, and soil type) in terms of how strongly they affect the "species" composition of the bacterial communities.*

Bacteria in the Nitrogen Cycle

Because nitrogen leaches easily from soils but is required in large amounts by plants, no mineral deficiency is more limiting to plant growth than a lack of nitrogen. The forms of nitrogen that plants can use include NO_3^- and NH_4^+. Some soil nitrogen derives from the weathering of rocks, and lightning produces small amounts of NO_3^- that get carried to the soil in rain. However, most of the nitrogen available to plants comes from the activity of bacteria **(Figure 37.11)**. This activity is part of the **nitrogen cycle**, a series of natural processes by which certain nitrogen-containing substances from the air and soil are made available to living things, are used by them, and are returned to the air and soil (see Figure 55.14).

Figure 37.11

The roles of soil bacteria in the nitrogen nutrition of plants.

The roles of soil bacteria in the nitrogen nutrition of plants.

N 2 is cycled between soil and atmosphere. In the soil, N 2 is processed by nitrogen-fixing bacteria that produce N H 3 (ammonia). H plus (from soil) is added to make N H 4 superscript plus (ammonium). Proteins from humus (dead organic material) go through microbial decomposition, break into amino acids, the ammonifying bacteria produce N H 4 superscript plus, as well. N H 4 superscript plus is taken up by roots, or nitrifying bacteria produce N O 2 superscript minus (nitrite) that is processed by nitrifying bacteria again to become N O 3 superscript minus (nitrate). Weathering of rock also produces this. Denitrifying bacteria then produce N 2 that is released from the soil. From the root, N H 4 superscript plus, nitrate, and nitrogenous organic composes are exported in the xylem to the shoot system.

Ammonium (NH_4^+) is made available to plants by two types of soil bacteria: those that fix atmospheric N_2 (nitrogen-fixing bacteria) and those that decompose organic material (ammonifying bacteria). Although plants absorb some ammonium from the soil, they absorb mainly nitrate, which is produced from ammonium by nitrifying bacteria. Plants reduce nitrate back to ammonium before incorporating the nitrogen into organic compounds.

VISUAL SKILLS *If an animal died near a root, would the plant have greater access to ammonium, nitrate, or both?*

*For suggested answer, see*Appendix A*.*

Figure 37.11 Walkthrough

The roles of soil bacteria in the nitrogen nutrition of plants.

Watch an explanation of this figure in the Figure 37.11 Walkthrough: The roles of soil bacteria in the nitrogen nutrition of plants.

image

Play

Plants commonly acquire nitrogen in the form of NO_3^- (nitrate). Soil NO_3^- is largely formed by a two-step process called *nitrification*, which consists of the oxidation of ammonia (NH_3) to nitrite (NO_2^-), followed by oxidation of NO_2^- to NO_3^-. Different types of *nitrifying bacteria* mediate each step, as shown at the bottom of Figure 37.11. After the roots absorb NO_3^-, a plant enzyme reduces it back to NH_4^+, which other enzymes incorporate into amino acids and other organic compounds. Most plant species transport nitrogen from roots to shoots via the xylem as NO_3^- or as organic compounds synthesized in the roots. Some soil nitrogen is lost, particularly in anaerobic soils, when denitrifying bacteria convert NO_3^- to N_2, which diffuses into the atmosphere.

In addition to NO_3^-, plants can acquire nitrogen in the form of NH_4^+ (ammonium) through two processes, as shown on the left in Figure 37.11. In one process, *nitrogen-fixing bacteria* convert gaseous nitrogen (N_2) to NH_3, which then picks up another H^+ in the soil solution, forming NH_4^+. In the other process, called ammonification, decomposers convert the organic nitrogen from dead organic material into NH_4^+.

Bacteria and Nitrogen Fixation

Although Earth's atmosphere is 79% nitrogen, plants cannot use free gaseous nitrogen (N_2) because there is a triple bond between the two nitrogen atoms, making the molecule almost inert. For atmospheric N_2 to be of use to plants, it must be reduced to NH_3 by a process known as ****__nitrogen fixation__. All nitrogen-fixing organisms are bacteria. Some nitrogen-fixing bacteria are free-living in the soil (see Figure 37.11), whereas others live in the rhizosphere. Among this latter group, members of the genus *Rhizobium* form efficient and intimate associations with the roots of legumes (such as beans, alfalfa, and peanuts), altering the structure of the hosts' roots markedly, as will be discussed shortly.

The multistep conversion of N_2 to NH_3 by nitrogen fixation can be summarized as follows:

$$N_2 + 8e^- + 8H^+ + 16ATP \rightarrow 2NH_3 + H_2 + 16ADP + 16Ⓟi$$

The reaction is driven by the enzyme complex *nitrogenase*. Because the process of nitrogen fixation requires 16 ATP molecules for every $2NH_3$ molecules synthesized, nitrogen-fixing bacteria require a rich supply of carbohydrates from decaying material, root secretions, or (in the case of the *Rhizobium* bacteria) the vascular tissue of roots.

The mutualism between *Rhizobium* ("root-living") bacteria and legume roots involves dramatic changes in root structure. Along a legume's roots are swellings called **nodules**, composed of plant cells "infected" by *Rhizobium* bacteria **(Figure 37.12)**. Inside each nodule, *Rhizobium* bacteria assume a form called **bacteroids**, which are contained within vesicles formed in the root cells. Legume-*Rhizobium* relationships generate more usable nitrogen for plants than all industrial fertilizers used today—and at virtually no cost to the farmer.

Figure 37.12

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Root nodules on a legume.
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Fluorescent LM of green bacteria on surface of root.

The spherical structures along this soybean root system are nodules containing Rhizobium bacteria. The bacteria fix nitrogen and obtain photosynthetic products supplied by the plant.

How is the relationship between legume plants and Rhizobium *bacteria mutualistic?*

*For suggested answer, see*Appendix A.**

Nitrogen fixation by *Rhizobium* requires an anaerobic environment, a condition facilitated by the location of the bacteroids inside living cells in the root cortex. The woody external layers of root nodules also help to limit gas exchange. Some root nodules appear reddish because of a molecule called leghemoglobin (*leg*- for "legume"), an iron-containing protein that binds reversibly to oxygen (similar to the hemoglobin in human red blood cells). This protein is an oxygen "buffer," reducing the concentration of free oxygen and thereby providing an anaerobic environment for nitrogen fixation while regulating the oxygen supply for the intense cellular respiration required to produce ATP for nitrogen fixation.

Each legume species is associated with a strain of *Rhizobium* bacteria. **Figure 37.13** describes how a root nodule develops after bacteria enter through an "infection thread" in a root hair. The symbiotic relationship between a legume and nitrogen-fixing bacteria is mutualistic. The bacteria supply the host plant with fixed nitrogen while the plant provides the bacteria with carbohydrates and other organic compounds. The root nodules use most of the ammonium produced to make amino acids, which are then transported up to the shoot through the xylem.

Figure 37.13

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Development of a soybean root nodule.
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Development of soybean root nodule.

1. Roots emit chemical signals that attract Rhizobium bacteria. The bacteria then emit signals that stimulate root hairs to elongate and to form an infection thread by an invagination of the plasma membrane.
2. The infection thread containing bacteria penetrates the root cortex. Cells of the cortex (outer root layer) and pericycle (just inside the endodermis) begin dividing, and vesicles containing the bacteria bud into cortical cells from the branching infection thread. Bacteria within the vesicles develop into nitrogen-fixing bacteriods.
3. Growth continues in the affected regions of the cortex and pericycle, and these two masses of dividing cells fuse, forming the nodule. The original infected root hair is sloughed off.
4. The nodule develops vascular tissue that supplies nutrients to the nodule and carries nitrogenous compounds into the vascular cylinder for distribution throughout the plant. The bacteriod continues to grow and bulges outside the root.

5. The mature nodule grows to be many times the diameter of the root. A layer of lignin-rich sclerenchyma cells forms around the nodule, reducing absorption of oxygen and thereby helping maintain the anaerobic environment needed for nitrogen fixation.

VISUAL SKILLS *What plant tissue systems are modified by root nodule formation?*

*For suggested answer, see*Appendix A.**

How does a legume species recognize a certain strain of *Rhizobium*? And how does an encounter with that specific *Rhizobium* strain lead to development of a nodule? Each partner responds to chemical signals from the other by expressing certain genes whose products contribute to nodule formation. By understanding the formation of root nodules, researchers hope to learn how to induce *Rhizobium* uptake and nodule formation in crop plants that do not normally form such nitrogen-fixing mutualistic relationships.

Nitrogen Fixation and Agriculture

The benefits of nitrogen fixation underlie most types of **crop rotation**. In this practice, a nonlegume such as maize is planted one year, and the following year alfalfa or some other legume is planted to restore the concentration of fixed nitrogen in the soil. To ensure that the legume encounters its specific *Rhizobium* strain, the seeds are exposed to bacteria before sowing. Instead of being harvested, the legume crop is often plowed under so that it will decompose as “green manure,” reducing the need for manufactured fertilizers.

Many plant families besides legumes include species that benefit from mutualistic nitrogen fixation. For example, red alder (*Alnus rubra*) trees host nitrogen-fixing actinomycete bacteria. Rice, a crop of great commercial importance, benefits indirectly from mutualistic nitrogen fixation. Rice farmers culture a free-floating aquatic fern, *Azolla*, which has mutualistic cyanobacteria that fix nitrogen (see the bottom of Figure 37.9). The growing rice eventually shades and kills the *Azolla*, and decomposition of this nitrogen-rich organic material increases the rice paddy’s fertility. Ducks also eat the *Azolla*, providing the rice paddy with an extra source of manure and providing the rice farmers with a source of meat.

Fungi and Plant Nutrition

Certain species of soil fungi also form mutualistic relationships with roots and play a major role in plant nutrition. Some of these fungi are endophytic, but the most important relationships are ****mycorrhizae** (“fungus roots”), the intimate mutualistic associations of roots and fungi (see Figure 31.4b). The host plant provides the fungus with a steady supply of sugar. Meanwhile, the fungus increases the surface area for water uptake and also supplies the plant with phosphorus and other minerals absorbed from the soil. The

fungi of mycorrhizae also secrete growth factors that stimulate roots to grow and branch, as well as antibiotics that help protect the plant from soil pathogens.

Mycorrhizae and Plant Evolution

EVOLUTION Mycorrhizae are not oddities; they are formed by most plant species. In fact, this plant-fungus mutualism might have been one of the evolutionary adaptations that helped plants initially colonize land (see Concept 29.1). When the earliest plants, which evolved from green algae, began to invade the land 400 to 500 million years ago, they encountered a harsh environment. Although the soil contained mineral nutrients, it lacked organic matter. Therefore, rain probably quickly leached away many of the soluble mineral nutrients. The barren land, however, was also a place of opportunities because light and carbon dioxide were abundant, and there was little competition or herbivory.

Neither the early land plants nor early land fungi were fully equipped to exploit the terrestrial environment. The early plants lacked the ability to extract essential nutrients from the soil, while the fungi were unable to manufacture carbohydrates. Instead of the fungi becoming parasitic on the rhizoids of the evolving plants (roots or root hairs had not yet evolved), the two organisms formed mycorrhizal associations, a mutualistic symbiosis that allowed both of them to exploit the terrestrial environment. Fossil evidence supports the idea that mycorrhizal associations occurred in the earliest land plants. The small minority of extant angiosperms that are nonmycorrhizal probably lost this ability through gene loss.

Types of Mycorrhizae

Mycorrhizae come in two forms called ectomycorrhizae and arbuscular mycorrhizae. **Ectomycorrhizae** form a dense sheath, or mantle, of mycelia (mass of branching hyphae) over the *surface* of the root. Fungal hyphae extend from the mantle into the soil, greatly increasing the surface area for water and mineral absorption. Hyphae also grow into the root cortex. These hyphae do not penetrate the root cells but form a network in the apoplast, or extracellular space, which facilitates nutrient exchange between the fungus and the plant. Compared with “uninfected” roots, the roots of ectomycorrhizae are generally thicker, shorter, and more branched. They typically do not form root hairs, which would be superfluous given the extensive surface area of the fungal mycelium. Only about 10% of plant families have species that form ectomycorrhizae. The vast majority of these species are woody, including members of the pine, oak, birch, and eucalyptus families.

Unlike ectomycorrhizae, **arbuscular mycorrhizae** (also called endomycorrhizae) do not ensheath the root but are embedded within it. They start when microscopic soil hyphae respond to the presence of a root by growing toward it, establishing contact, and growing along its surface. The hyphae penetrate between epidermal cells and then enter the root cortex, where they digest small patches of the cell walls but don’t

pierce the plasma membrane. Instead of entering the cytoplasm, a hypha grows into a tube formed by invagination of the root cell's membrane. This invagination is like poking a finger gently into a balloon without popping it; your finger is like the fungal hypha, and the balloon skin is like the root cell's membrane. After the hyphae have penetrated in this way, some of them branch densely, forming structures called arbuscules ("little trees"), which are important sites of nutrient transfer between the fungus and the plant. Within the hyphae themselves, oval vesicles may form, possibly serving as food storage sites for the fungus. Arbuscular mycorrhizae are far more common than ectomycorrhizae, being found in over 85% of plant species, including most crops. About 5% of plant species don't form mycorrhizal associations. **Figure 37.14** provides an overview of mycorrhizae.

Figure 37.14

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Mycorrhizae.
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Mycorrhizae.

Ectomycorrhizae. The mantle of the fungal mycelium ensheathes the root. Fungal hyphae extend from the mantle into the soil, absorbing water and minerals, especially phosphorus. Hyphae also extend into the extracellular spaces of the root cortex, providing extensive surface area for nutrient exchange between the fungus and its host plant. Colorized SEM of the mantle (fungal sheath). Scale: 1.5 millimeters. Exploded cross-section of the root showing that the mantle ensheathes the root and extends into the extracellular spaces of the root cortex but does not penetrate the cell walls. LM view of the fungal hyphae between cortical cells (Scale: 50 micrometers). Arbuscular mycorrhizae (endomycorrhizae). No mantle forms around the root, but microscopic fungal hyphae extend into the root. Within the root cortex, the fungus makes extensive contact with the plant through branching of hyphae that form arbuscules, providing an enormous surface area for nutrient swapping. The hyphae penetrate the cell walls, but not the plasma membranes of cells within the cortex. Exploded cross-section of the root showing that no mantle forms around the root but that the fungal hyphae surround the root and extend into the root and penetrate the cell walls, but not the plasma membranes. LM view of the cortical cell and arbuscules (Scale: 10 micrometers).

Agricultural and Ecological Importance of Mycorrhizae

Good crop yields often depend on the formation of mycorrhizae. Roots can form mycorrhizal symbioses only if exposed to the appropriate species of fungus. In most ecosystems, these fungi are present in the soil, and seedlings develop mycorrhizae. But if crop seeds are collected in one environment and planted in foreign soil, the plants may show signs of malnutrition (particularly phosphorus deficiency) resulting from the absence of fungal partners. Treating seeds with spores of mycorrhizal fungi can help seedlings form

mycorrhizae, facilitating recovery of damaged natural ecosystems (see Concept 55.5) and improving crop yield.

Mycorrhizal associations are also important in understanding ecological relationships. Arbuscular mycorrhizae fungi exhibit little host specificity; a single fungus may form a shared mycorrhizal network with several plants, even plants of different species. Mycorrhizal networks in a plant community may benefit one plant species more than another. Other examples of how mycorrhizae may affect the structures of plant communities come from studies of introduced plant species. For instance, garlic mustard (*Alliaria petiolata*), a European species that has been introduced into the forests of the eastern half of the United States, does not form mycorrhizae but hinders the growth of other plant species by preventing the growth of arbuscular mycorrhizal fungi.

Epiphytes, Parasitic Plants, and Carnivorous Plants

Almost all plant species have mutualistic relationships with soil fungi, bacteria, or both. Some plant species, including epiphytes, parasites, and carnivores, have unusual adaptations that facilitate exploiting other organisms **(Figure 37.15)**.

Figure 37.15

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Exploring Unusual Nutritional Adaptations in Plants
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Figure explores unusual nutritional adaptations in plants, such as epiphytes, parasitic plants, and carnivorous plants.

Epiphytes: An epiphyte (from the Greek epi, upon, and phyton, plant) is a plant that grows on another plant. Epiphytes produce and gather their own nutrients; they do not tap into their hosts for sustenance. Usually anchored to the branches or trunks of living trees, epiphytes absorb water and minerals from rain, mostly through leaves rather than roots. Some examples are staghorn ferns, bromeliads, and many orchids, including the vanilla plant. Photo shows a staghorn fern (an epiphyte) growing around a tree.

Parasitic plants: Unlike epiphytes, parasitic plants absorb water, minerals, and sometimes products of photosynthesis from their living hosts. Many species have roots that function as haustoria, nutrient-absorbing projections that tap into the host plant. Some parasitic species, such as orange-colored, spaghetti-like dodder (genus Cuscuta), lack chlorophyll entirely, whereas others, such as mistletoe (genus Phoradendron), are photosynthetic. Still others, such as Indian pipe (Monotropa uniflora), absorb nutrients from the hyphae of mycorrhizae associated with other plants. Photos show dodder (a nonphotosynthetic

parasite which looks like an orange band wrapped around its host plant) and Indian pipe (a nonphotosynthetic whitish parasite of mycorrhizae).

Carnivorous plants: Carnivorous plants are photosynthetic but supplement their mineral diet by capturing insects and other small animals. They live in acid bogs and other habitats where soils are poor in nitrogen and other minerals. Pitcher plants such as Nepenthes and Sarracenia have water-filled funnels into which prey slip and drown, eventually to be digested by enzymes. Sundews (genus Drosera) exude a sticky fluid from tentacle-like glands on highly modified leaves. Stalked glands secrete sweet mucilage that attracts and ensnares insects, and they also release digestive enzymes. Other glands then absorb the nutrient soup. The highly modified leaves of Venus flytrap (Dionaea muscipula) close quickly but partially when a prey hits two trigger hairs in rapid enough succession. Smaller insects can escape, but larger ones are trapped by the teeth lining the margins of the lobes. Excitation by the prey causes the trap to narrow more and digestive enzymes to be released. Photos show pitcher plants, Venus flytraps, and Sundew.

A recent study suggests that exploiting other organisms may be the norm. Chanyarat Paungfoo-Lonhienne and her colleagues at the University of Queensland in Australia have provided evidence that *Arabidopsis* and tomato can take up bacteria and yeast into their roots and digest them. Due to the small pore size of the cell wall (less than 10nm) relative to the size of bacterial cells (about 1,000nm), taking in microorganisms may depend on digestion of the cell wall. A study with wheat suggests that microorganisms provide only a tiny fraction of the plant's nitrogen needs, but this may not be true for all plants. These findings suggest that many plant species might engage in a limited amount of carnivory.